

Jaa Jong Tee

Jong Jing Jung Company

**Mobile reservation app that simplifies seat booking and
pre-ordering**

Project Phase #2

By

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**ITCS224 Fundamentals of Information Systems & Software
Development**

**Faculty of Information and Communication Technology
Mahidol University**

2024

Introduction of Jaa Jong Tee

Overview of Jaa Jong Tee

Jaa Jong Tee is a technology-driven company that is dedicated to revolutionizing the dining and café experience by addressing critical challenges such as seating availability, long wait times, and inefficient service management.

Our flagship product, the reservation website, provides a comprehensive solution for both customers and restaurants by enabling seat reservations, pre-ordering food and beverages, and seamless digital payments. Customers can book a seat ahead of time, select their meals, pay for their meals, and arrive at a venue knowing their order is ready. This eliminates peak-hour frustration, disorganized seating arrangements, and long wait times, ensuring a smooth and stress-free dining experience.

Beyond individual diners, Ja Jong Tee also supports business professionals, students, and remote workers who need reliable seating and work-friendly environments in cafés and restaurants. The app allows business groups to book meeting spaces, providing a platform for corporate events, networking gatherings, and team discussions.

With its real-time availability tracking, the platform helps restaurants and cafés optimize seat utilization and service efficiency while ensuring a personalized, hassle-free experience for customers.

Target customer

1. Restaurants and Cafes (B2B)

Restaurants and cafes, establishments looking to improve customer flow, increase revenue, and enhance overall customer satisfaction. Particularly those in high-traffic urban areas, are an essential part of the app's target market. The app enables businesses to manage reservations efficiently, reduce wait times, and optimize seating arrangements. It also helps them handle peak-hour demand, increasing customer satisfaction while improving operational efficiency.

2. Students Looking for Study Spaces

Many students face difficulty finding a quiet, comfortable space to do homework or read books, especially during peak hours. The app offers students a solution by allowing them to reserve seats in cafes or restaurants where they can work, study, and read without the stress of finding a seat. This ensures a conducive study environment away from home.

3. Professionals Seeking a Flexible Workspace & Business Meetings

Modern professionals, including remote workers, flexible workspace, digital nomads, and corporate teams, often need a comfortable and productive workspace outside the office. Whether working individually or conducting business meetings, team discussions, or networking events, finding a suitable venue with available seating can be a challenge. The Mobile Reservation App caters to these needs by allowing users to reserve seats in cafes and restaurants that offer ideal workspaces. Individuals can secure quiet seating to focus on tasks while enjoying coffee or a meal, while business groups can book private or group spaces for meetings, corporate gatherings, or networking sessions. This ensures a professional yet relaxed setting with pre-ordered refreshments and seamless service, making work and collaboration more efficient.

Definition of system

To use the website, the customer needs to have an account. For each customer to have their account, they have to create an account for themselves. An account can be created by getting the information from the customer. The user has to give their personal information to the system via the sign-up page, such as first name, last name, phone number, etc. Then, the customer can log in to the website via the login page. The customer must log in to the website with their existing account.

Every customer gets a chance to use a coupon code for their first order, which is provided by the system, since they are a new customer. Each coupon code can be used only once by each user. The customer has to click the coupon code button to use the coupon for a discount on their order in their selected coffee shop or restaurant.

The customer can place their order for food and drinks in their selected coffee shop or restaurant through the website. Firstly, after the customer has logged in to the website with their account. The customer will be able to select their coffee shop or restaurant of choice. Then, the user will have to select their food or drink, or both. Then, the customer will have a chance to select and reserve their seat in the chosen coffee shop or restaurant.

After the customer places their order for food and drinks in their selected coffee shop or restaurant, the user has to pay for their order. For the customer to pay their payment, the payment will be created by the system, which will use the information taken from the customer's coffee shop or restaurant of choice. The system will provide choices of methods of payment for the user. After the customer has selected their method of payment, they will have to pay with it. If the chosen method is the card, the user must give their card information to the system. If the chosen payment method is an online transaction, the QR code will be provided to the user by the system.

The system will notify the user's coffee shop or restaurant of choice of what they have ordered, including their reserved seat. After the system gives information to the user's coffee shop or restaurant of choice, the system will take the confirmation from the user's coffee shop or restaurant of choice, whether they are ready, complete, or cancel to the customer.

The admin can create a new shop to their desire, by giving the shop's name, menu. Seat layout. And opening time to the system. The admin can also edit and delete shops.

System Requirement

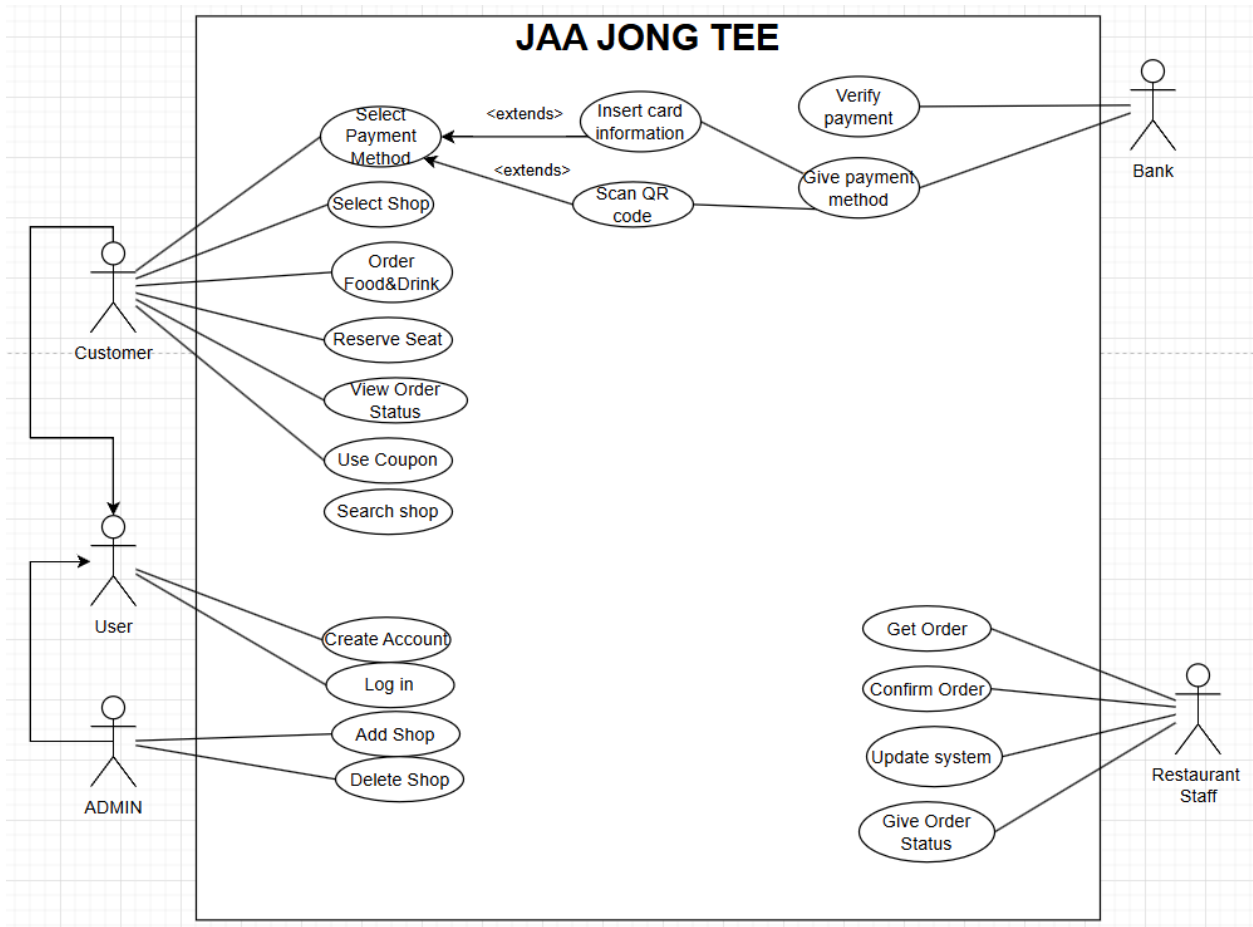
Functional Requirements

1. The user must be able to create an account by personal information.
2. Restaurant staff must be able to create and manage a staff account.
3. The user must be able to log in and log out of the system.
4. Restaurant staff must be able to log in and log out of the system.
5. The system provides a new user coupon code for each new user account.
6. The user must be able to view and select restaurants or cafes from the application.
7. The system displays relevant details such as name, menu, and available seats.
8. The user must be able to browse the menus for their selected restaurant.
9. The user must be able to select food or drinks for pre-order.
10. Restaurant staff must be able to view pre-orders made by users.
11. The user must be able to review and confirm their order and seat reservation.
12. The system must send the order and seat reservation to the restaurant staff.
13. The system must generate a bill based on the user's order.
14. The user must be able to choose a payment method, such as a credit card or QR code.
15. The system must notify the restaurant user and the restaurant staff.
16. Restaurant staff must receive real-time notifications for new orders and reservations.
17. The system must receive and display order status updates.
18. Restaurant staff must be able to update the status of each order.
19. The system must collect card details.
20. The system must allow users to reserve a seat at the restaurant.

Non-Functional Requirements

1. Real-time seat availability update within 2 seconds.
2. The system handles concurrent users at a time.
3. The system should be available 24/7, especially during peak hours
4. The system should provide a user navigation guide for first-time users.
5. The system should respond to the user within 5 seconds.
6. The reservation should be completed within 5 steps.
7. The application should be mobile-friendly.
8. The system should collect user data from over 500 users.
9. The system should automatically send the order status to the customer.
10. The search filter should work immediately after using it.

Use Case Diagram

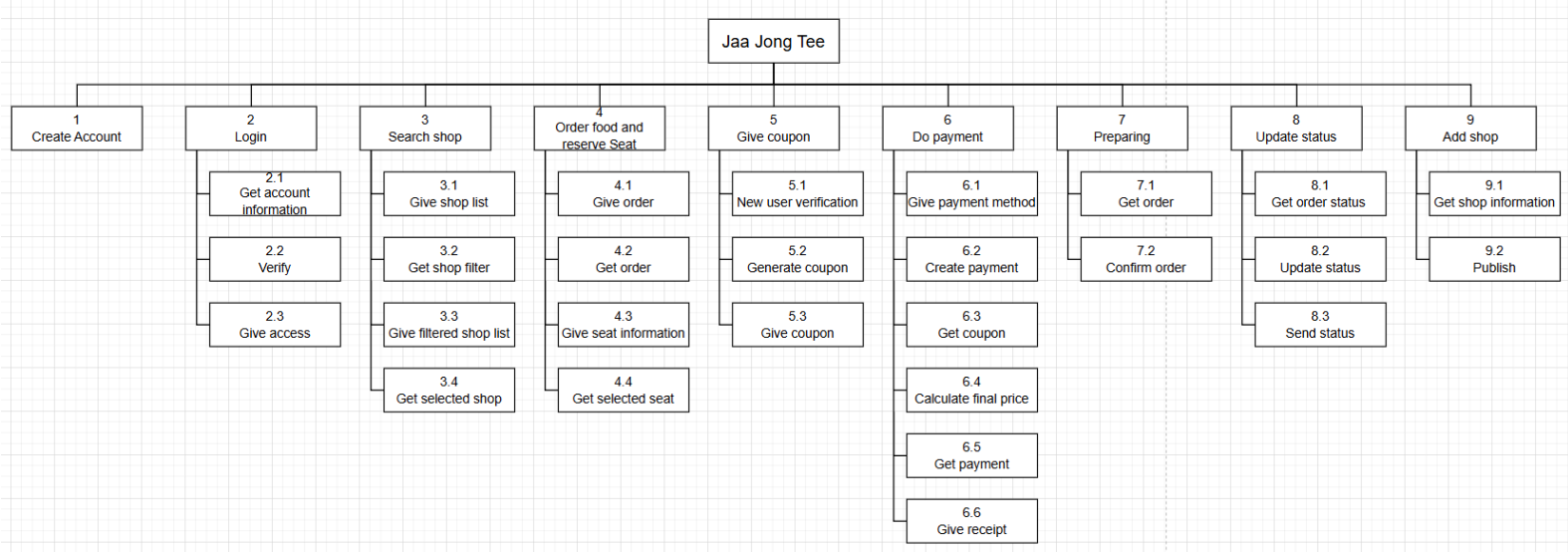


Use Case Narrative

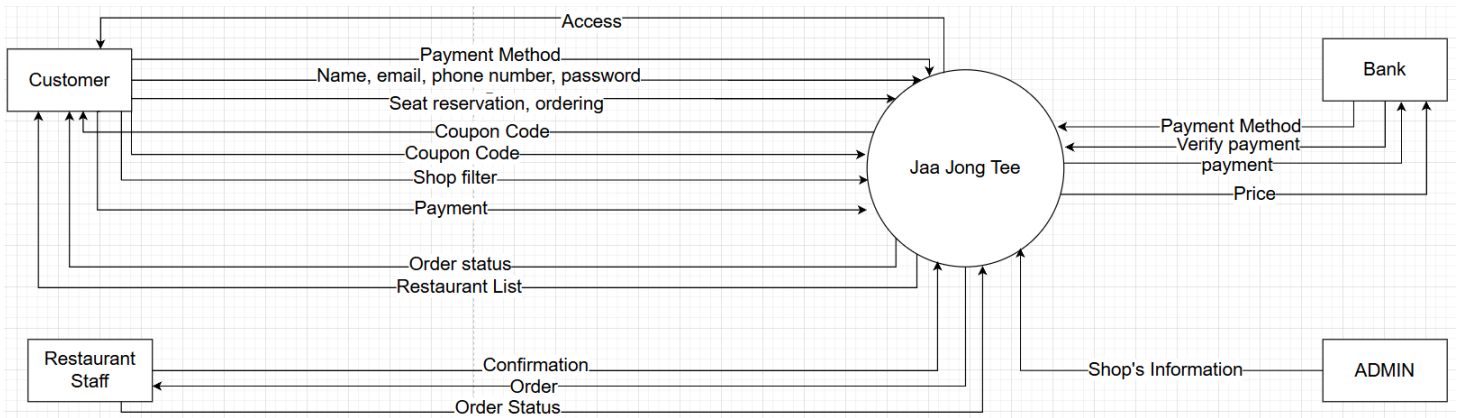
Use case Name	Reservation Application
Primary Actor	customer
Goal in Context	To be able to order food & drink and reserve a seat
Precondition	The customer needs to have an account.
Trigger	The customer wants to reserve a seat and order food.
Scenario	1. The customer browses Jaa Jong Tee's website 2. The website shows a main page. 3. The customer can choose the Login Page 4. Customer enters their email and Password 5. The website displays a homepage. 6. The customer starts to search for a shop. 7. The customer selects the cafe company, location, and opening hours. 8. The website shows restaurants. 9. The customer selects a restaurant preference. 10. The customer clicks "reserve now " under the restaurant image 11. The customer selects their order. 12. The customer selects seat preference, Date, and time, then clicks "continue".
Exception	1. Email or passwords are incorrect or not recognized - see use case: Validate phone number and passwords 2. No available restaurant - use the filter and select an available restaurant. 3. No available seat - select another seat preference.
Postcondition	Customers can reserve seats and order food via the website.

Use case Name	Reservation Application
Primary Actor	Customer
Goal in Context	Pay for their order.
Precondition	The customer selects their food, drink, and seat.
Trigger	The customer wants to finish the ordering process.
Scenario	1. The customer selects the payment method. 2. The customer adds their card information. 3. The customer chooses to use the coupon code. 4. The system calculates the final price with the coupon code. 5. The customer pays. 6. The system gives the receipt to the customer.
Exception	-
Postcondition	The order will be sent to the shop staff.

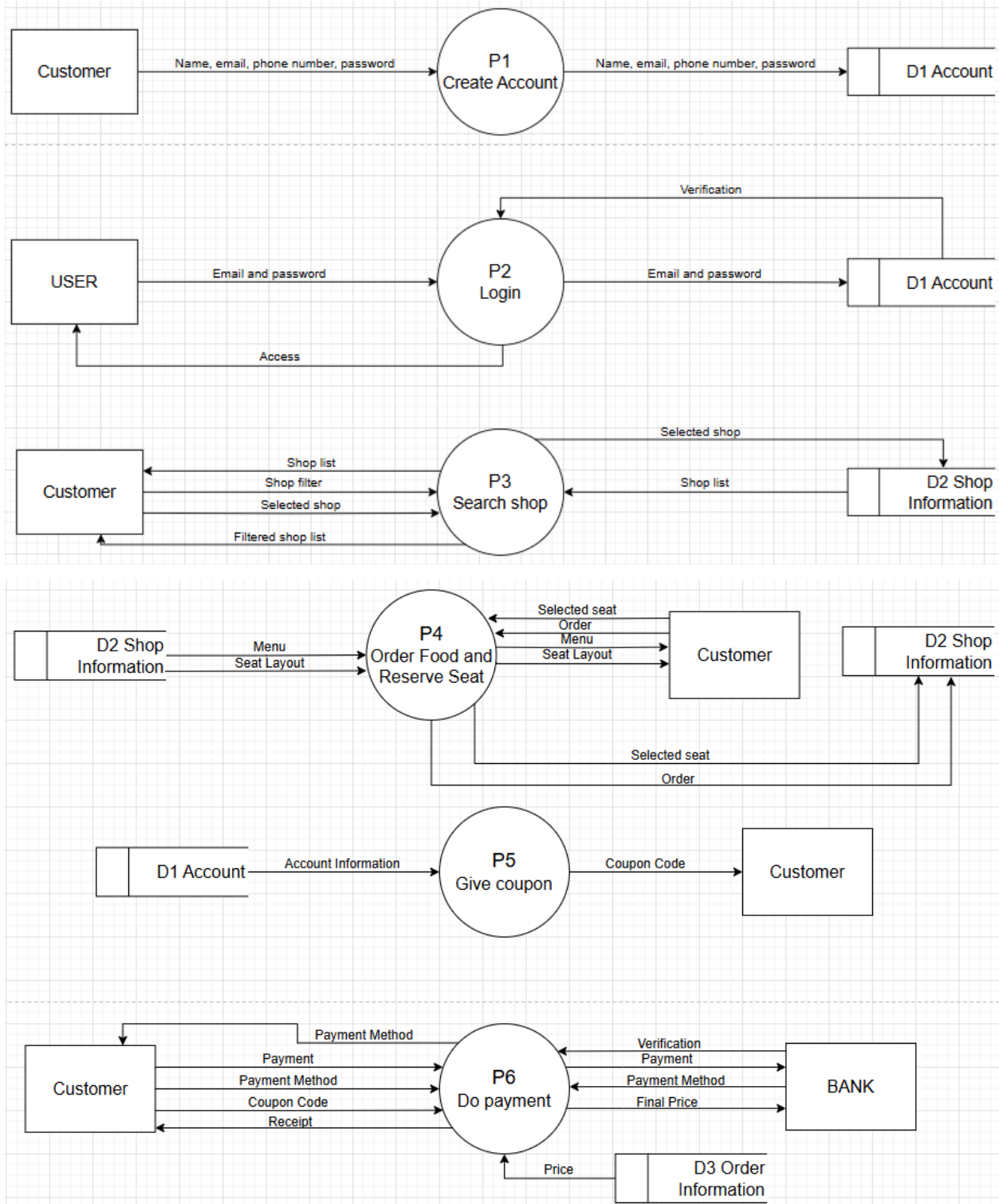
Functional Decomposition Diagram

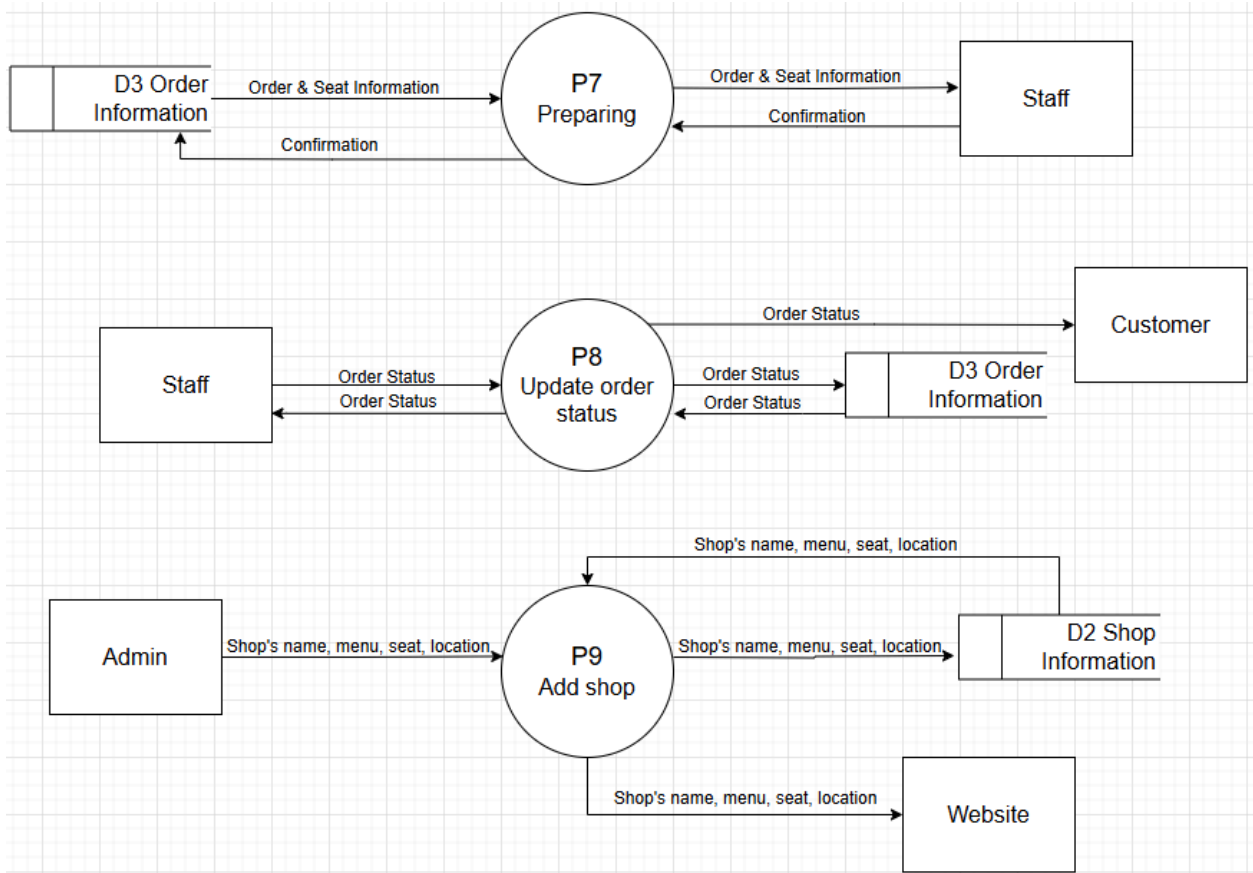


Data Flow Diagram Level 0(Context Diagram)



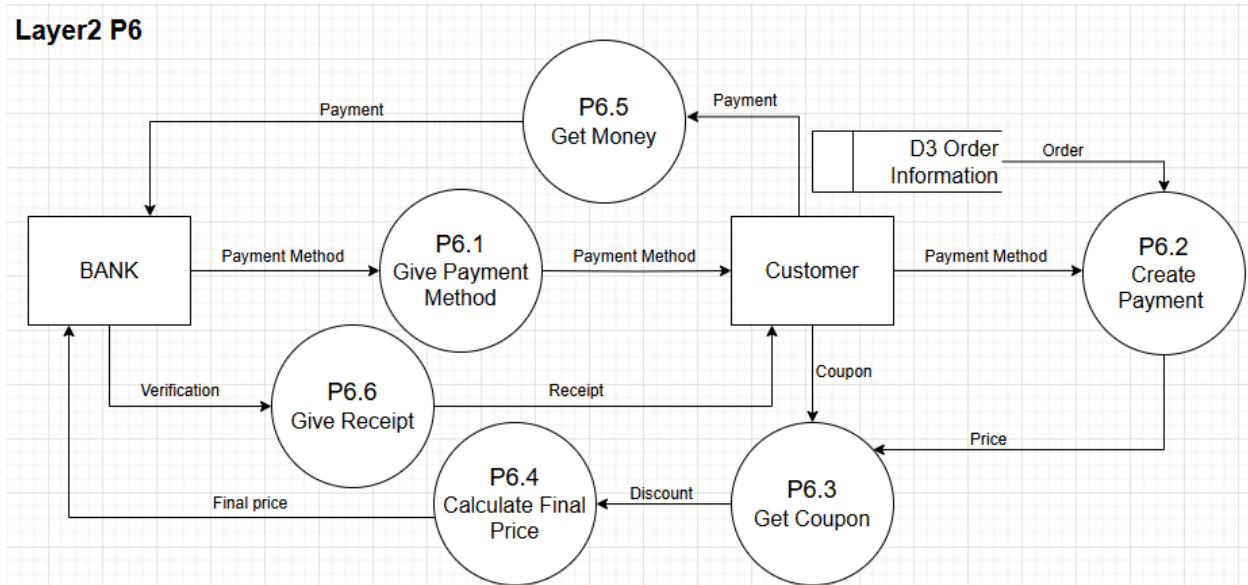
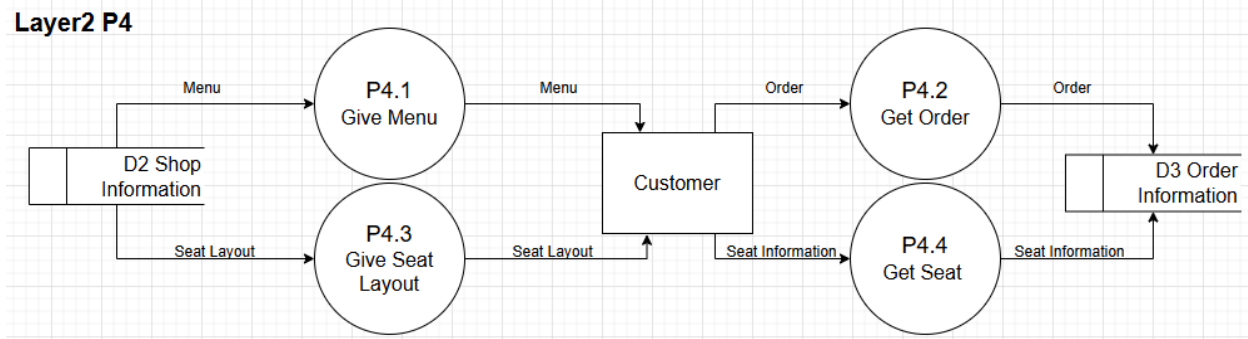
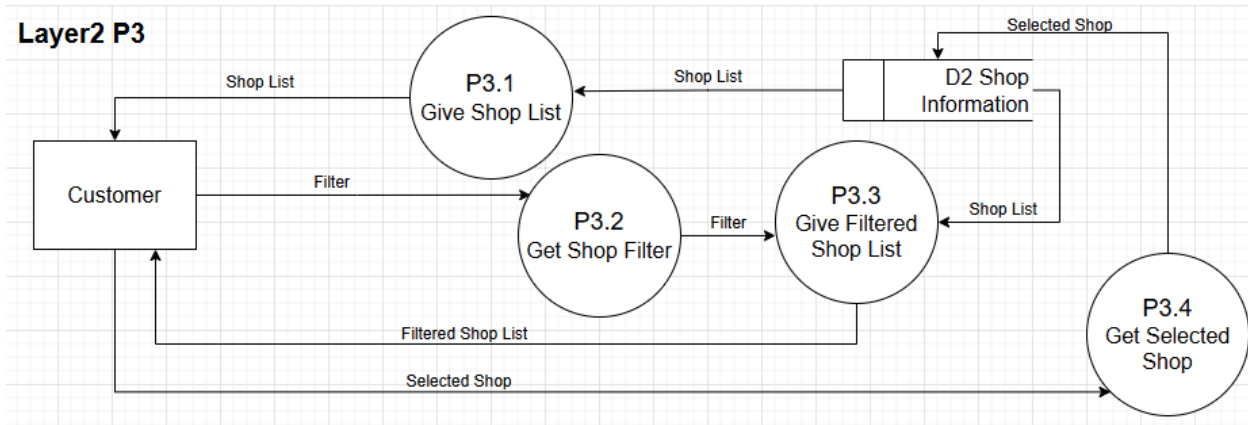
Data Flow Diagram Level 1





Data Flow Diagram Level 2

The three main processes that we chose to do in this report are **Process 3, Process 4, and Process 6**, which are **select shop, order and reservation, and payment** in order.



Prototype

[Figma Link](#)

Process 3

The image shows a web application prototype for 'JAA JONG T'. The top navigation bar is purple and contains the logo on the left, a search bar in the center, and three icons (notification, settings, and user profile) on the right. Below the navigation bar, the main content area is light gray. It features three filter sections on the left: 'Cafe' with a 'Select Cafe' dropdown menu, 'Location' with a 'Select Location' dropdown menu, and 'Opening Hours' with a 'Select Opening Hours' dropdown menu. To the right of these filters, there are two buttons: a purple 'Search' button and a gray 'Reset' button.

Search Page

On the top screen, the Website provides a search function for the users. The users can select or filter their restaurant preferences, such as Cafe company, Location, and Opening Hours.

JAA
JONG
T

Cafe

Select Cafe ^

Starbucks

Location

Amazon Cafe

Punthal Coffee

Opening Hours

Search

Reset

Select Cafe's brand; in this case, they selected Starbucks.

JAA
JONG
T

Cafe

Starbucks v

Location

Select Location ^

Near me

Bangkok

Nakhon Pathom

Opening Hours

Search

Reset

Select the location of the cafe; in this case, they selected Near me.

JAA
JONG
T

+

Cafe

Starbucks

Location

Near me

Opening Hours

Select Opening Hours

Standard Hours

Open 24 Hours

Search

Reset

And the last filter is the opening hours; in this case, we select Standard Hours.

JAA
JONG
T

+

Cafe

Starbucks

Location

Near me

Opening Hours

Standard Hours

Search

Reset

STARBUCKS

Mahidol University - Salaya Campus

999 Phutthamonthon 4 Road , Salaya , Nakhon
Patthom 73170

Monday 07:00 to 19:00

Tuesday 07:00 to 19:00

Wednesday 07:00 to 19:00

Thursday 07:00 to 19:00

Friday 07:00 to 19:00

Saturday 08:00 to 17:00

Sunday 08:00 to 17:00

The Fourth Putthamonthon Sai 4

Phutthamonthon Sai 4 Road ,Kratumliom,Sam
Phran,Nakornpathom

Monday 07:00 to 21:00

Tuesday 07:00 to 21:00

Wednesday 07:00 to 21:00

Thursday 07:00 to 21:00

Friday 07:00 to 21:00

Saturday Closed

Sunday Closed

Then the customer clicks search, the system will filter out shops and give us options of shops.

JAA
JONG
T

+



Cafe

Starbucks

Location

Near me

Opening Hours

Standard Hours

Search

Reset



Mahidol University - Salaya Campus

999 Phutthamonthon 4 Road , Salaya , Nakhon
Pathom 73170

Monday 07:00 to 19:00

Tuesday 07:00 to 19:00

Wednesday 07:00 to 19:00

Thursday 07:00 to 19:00

Friday 07:00 to 19:00

Saturday 08:00 to 17:00

Sunday 08:00 to 17:00



The Fourth Putthamonthon Sai 4

Phutthamonthon Sai 4 Road ,Kratumlon,Sam
Phran,Nakornpathom

Monday 07:00 to 21:00

Tuesday 07:00 to 21:00

Wednesday 07:00 to 21:00

Thursday 07:00 to 21:00

Friday 07:00 to 21:00

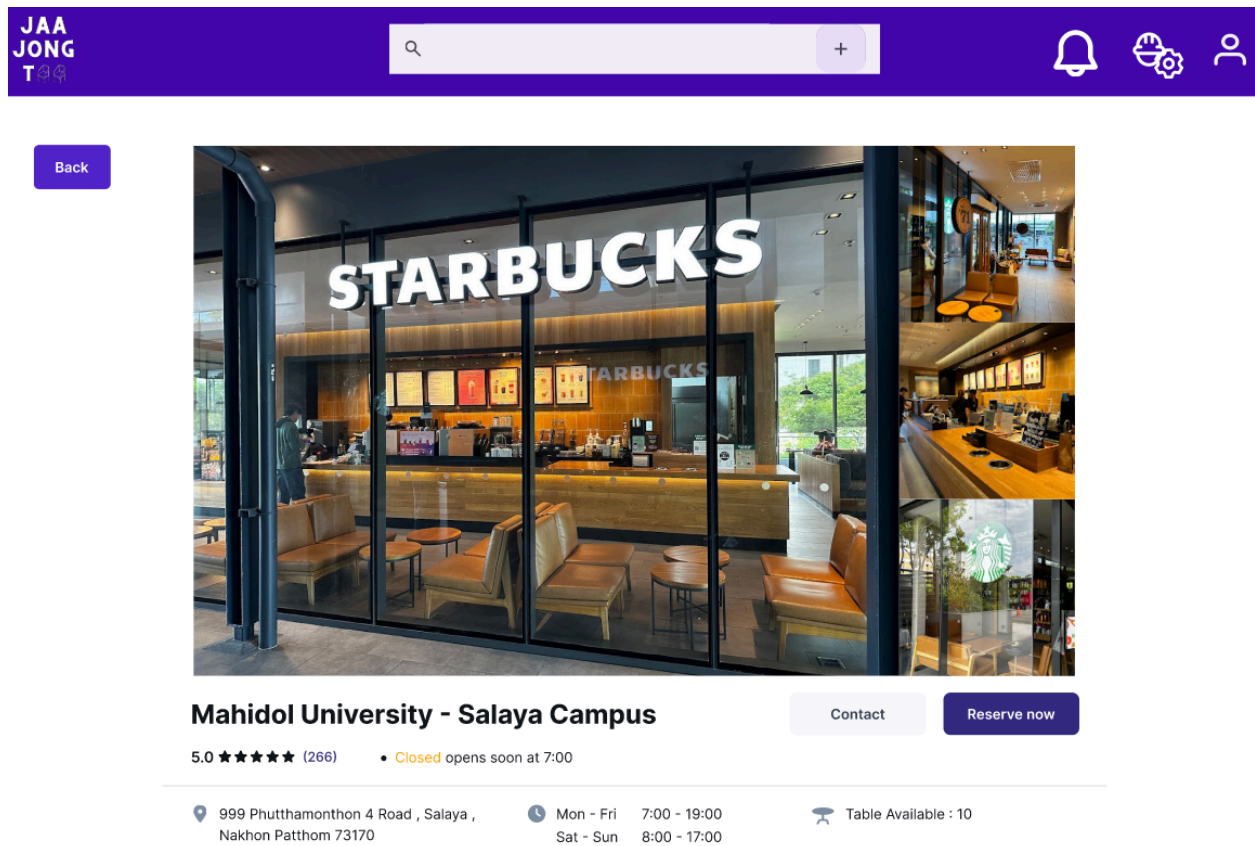
Saturday Closed

Sunday Closed



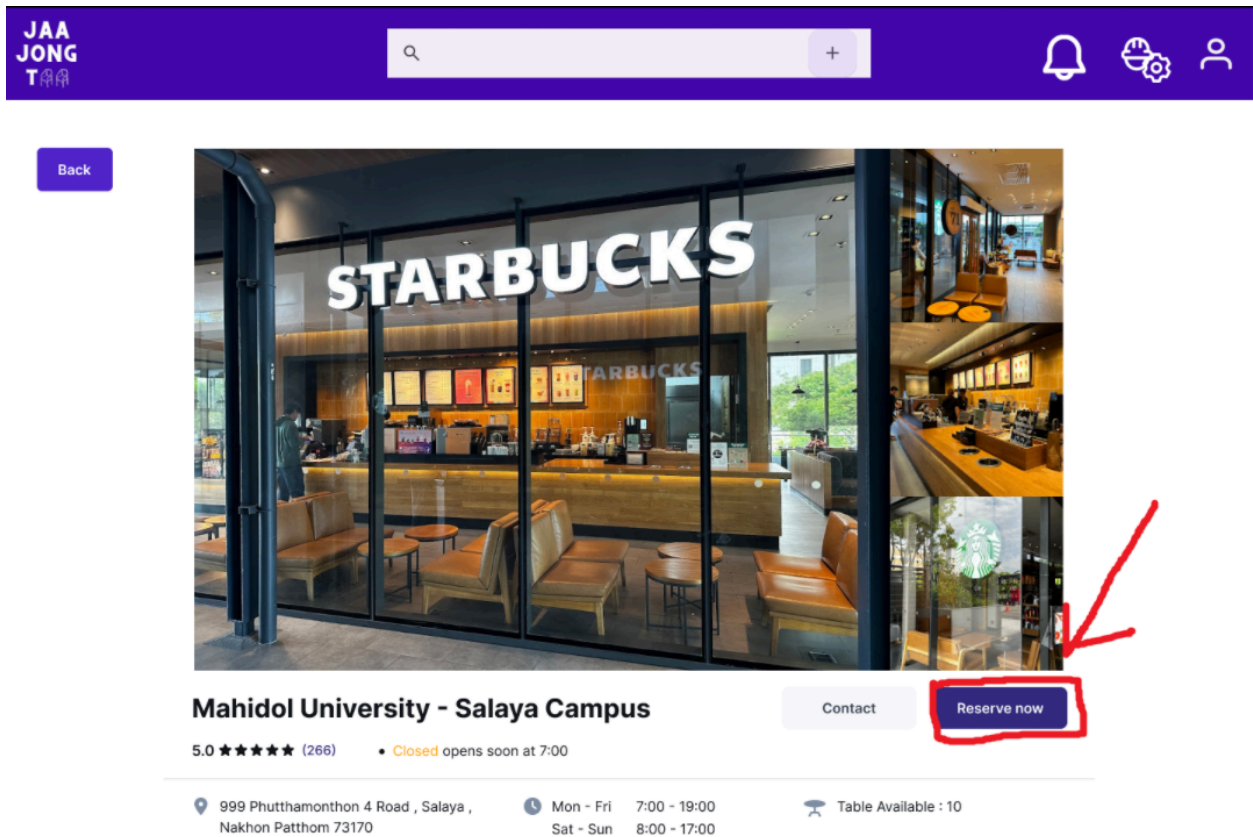
Then the customer can choose the show that they want.

Process 4

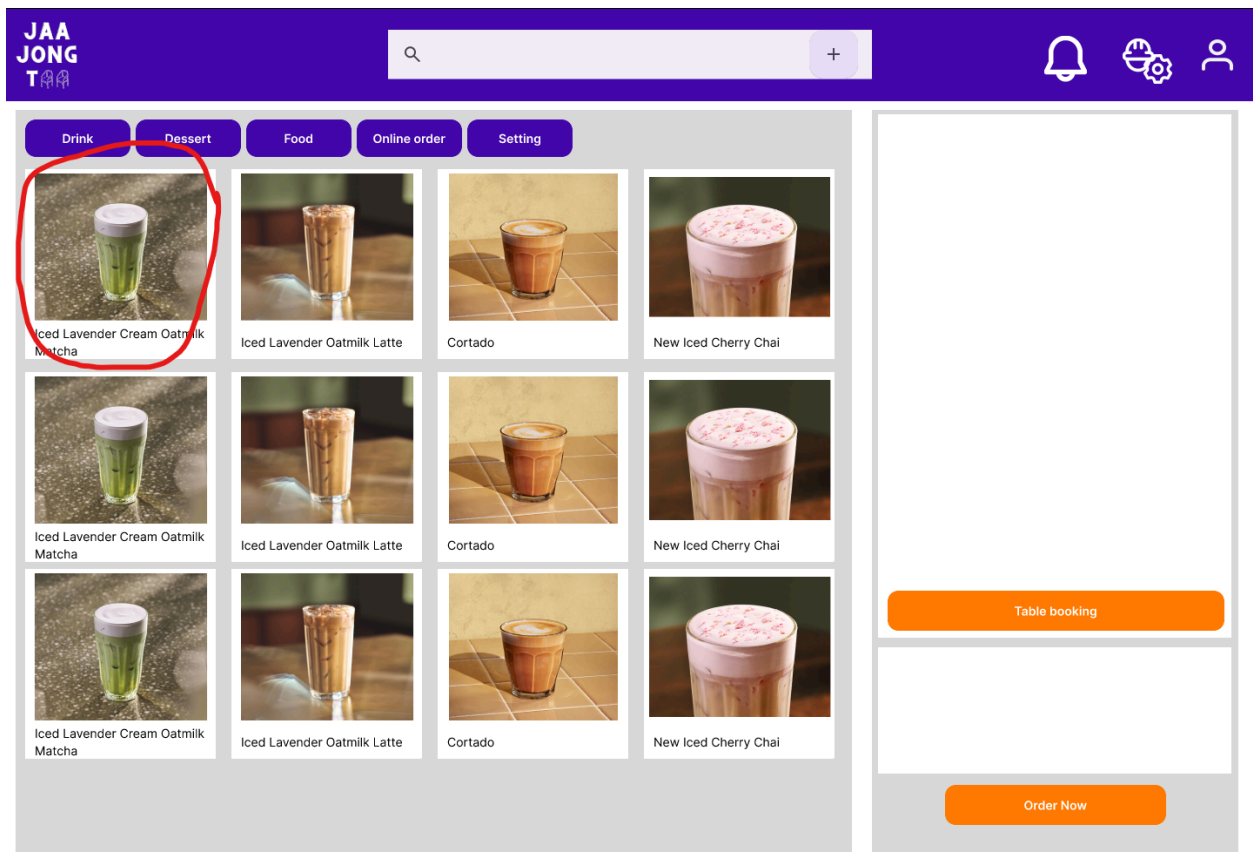


Restaurant Page

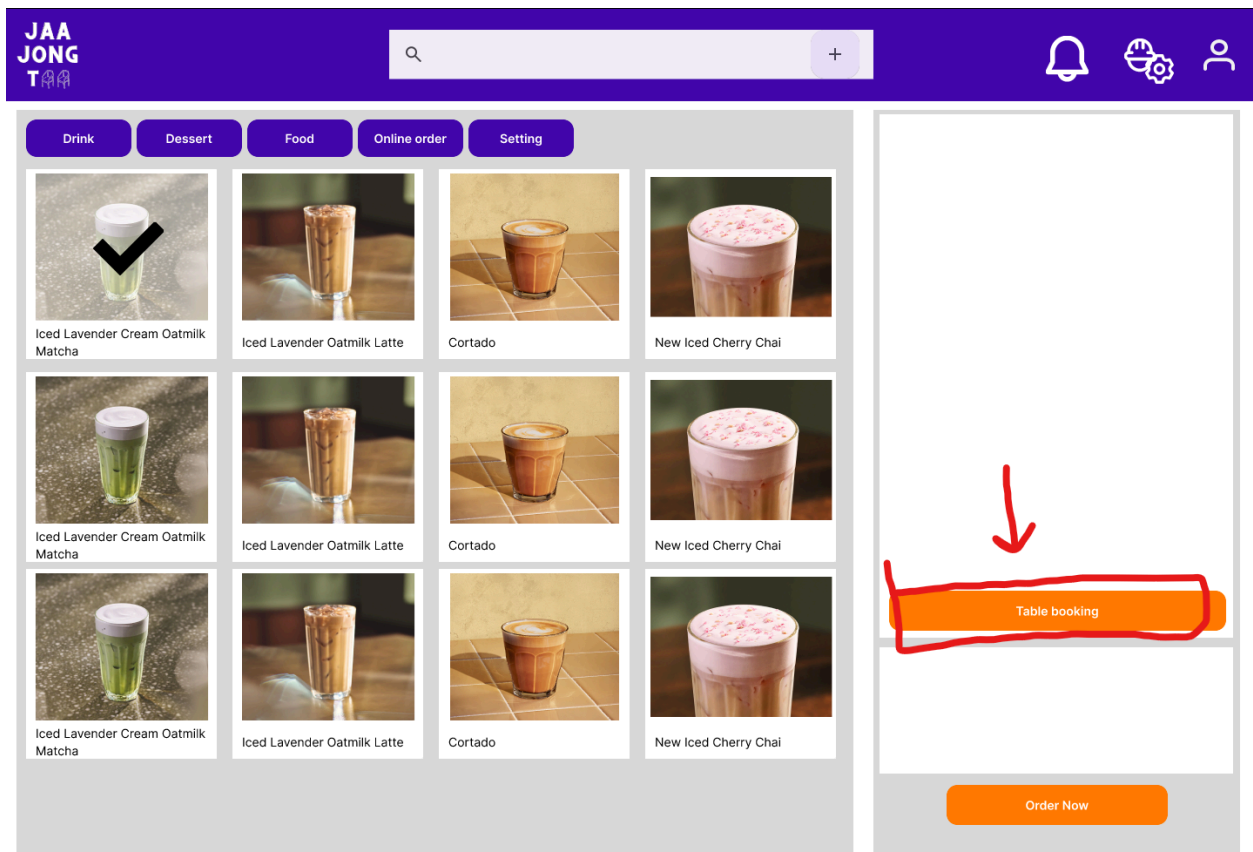
When the users select their restaurant, the website displays the restaurant's information such as Name, company, Location, Opening Hours, Table availability, and The User can reserve or contact their restaurant via the website.



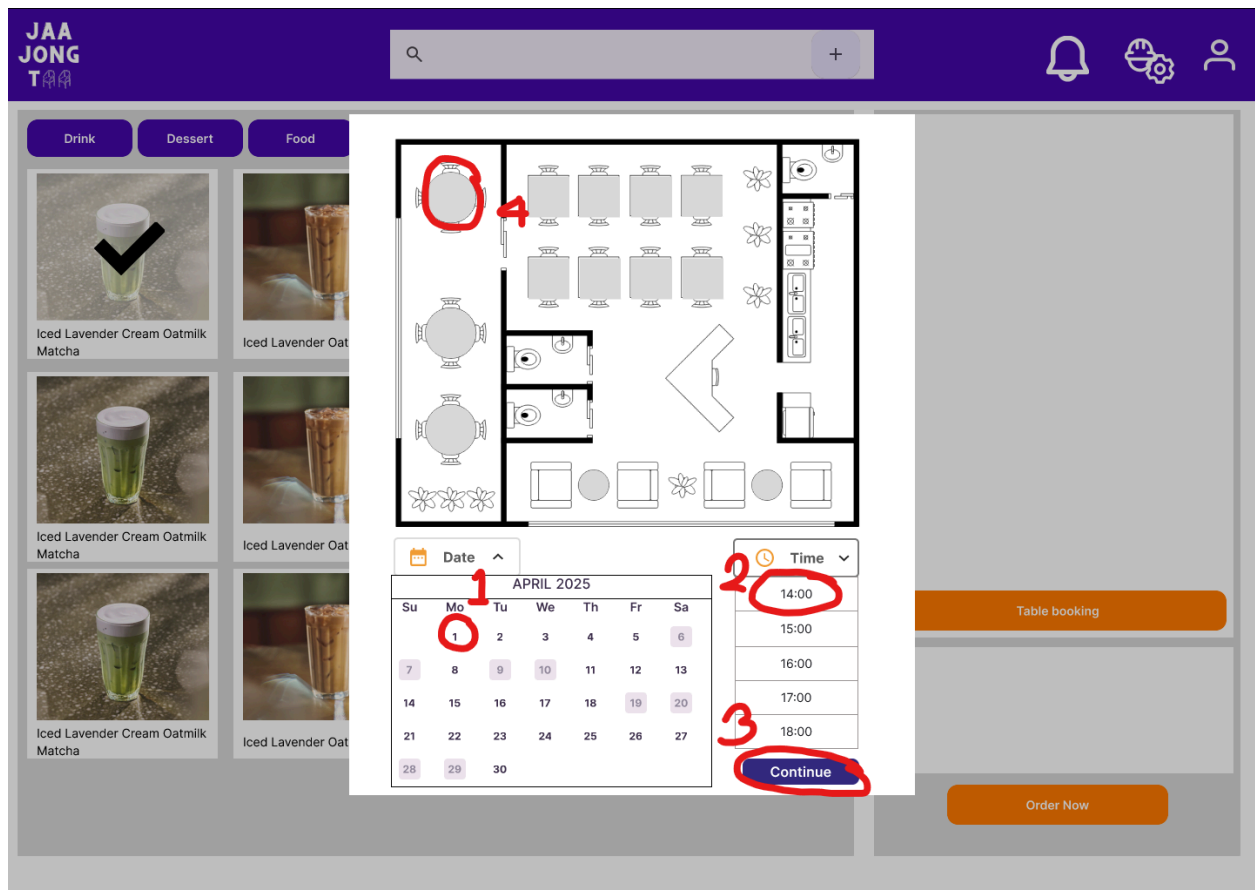
After the customer has filtered and chosen the shop, they can click “Reserve now”. The system will bring us to the order page.



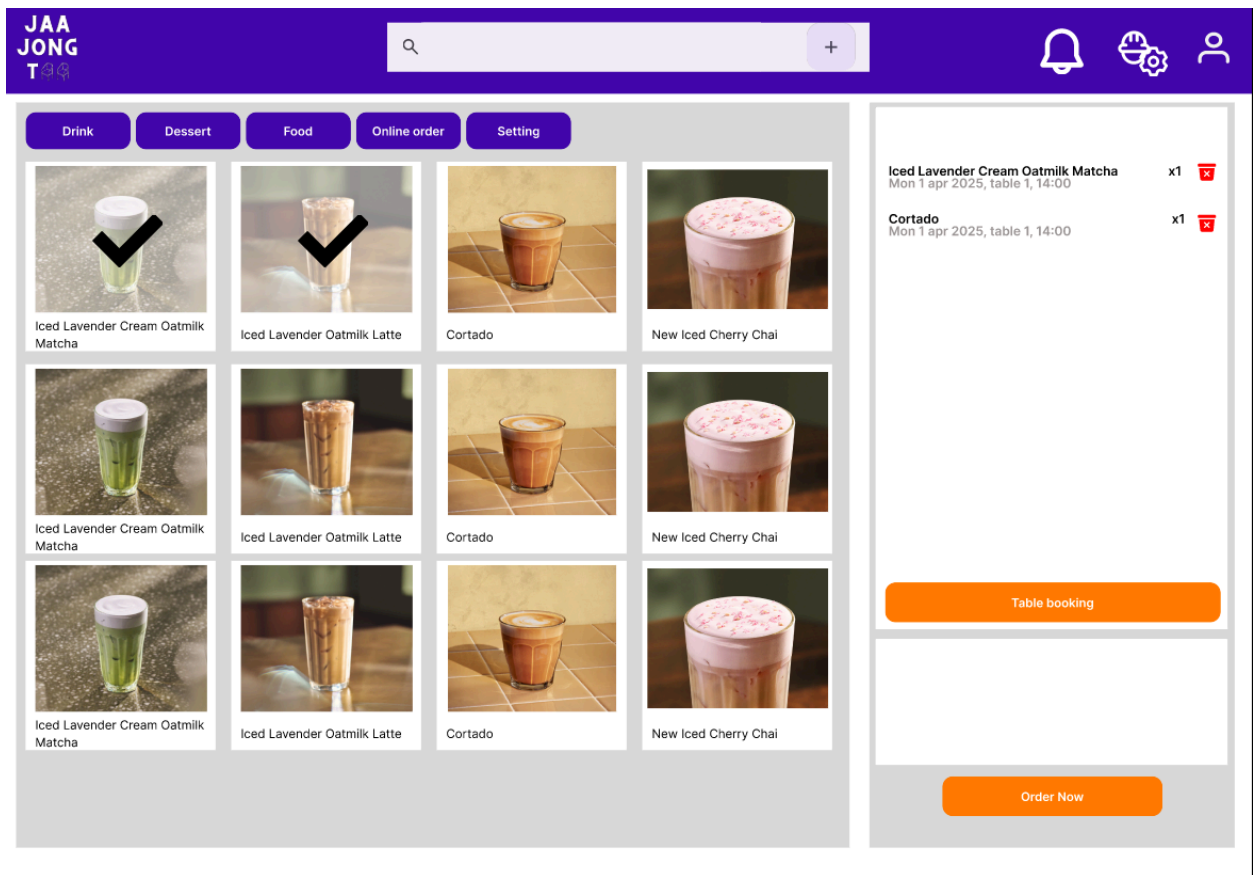
On this page, the customer can order their food and drinks by clicking on the food picture.



After the customer has chosen their food, they have to reserve their table by clicking the “Table booking” button.



The customer can reserve their specific table at a specific time by clicking on the following button. 1 is for Date, 2 is for Time, 3 is to continue, and 4 is to choose the specific table.



After clicking the continue button, the Website displays the user's order on the right screen, and the user can also delete their order or product at the red bin button. Then click the "Order Now" button.

Process 6

The screenshot shows the JAA JONG website interface. A modal form is overlaid on the product list, providing a checkout flow. The modal includes sections for payment method selection, payment details entry, and order confirmation. A red arrow highlights the transition from the product list to the modal's 'Order Now' button.

Payment Method Selection:

- Master Card
- Visa
- PayPal

Payment Details:

- Name
- Card number
- DD/MM/YYYY
- CVV CODE
- Coupon

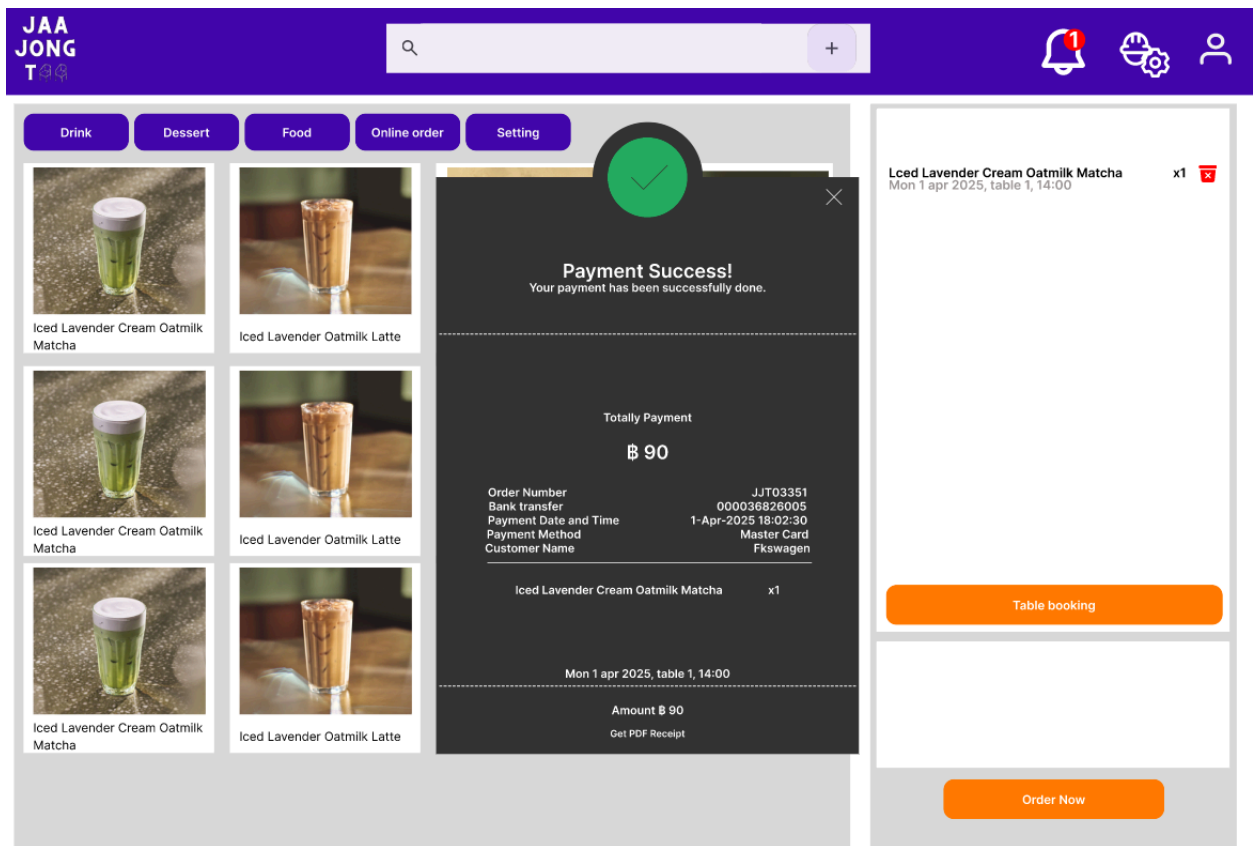
Order Confirmation:

- Starbucks logo
- Mahidol University - Salaya Campus
- 999 Phutthamonthon 4 Road, Salaya, Nakhon Pathom 73170
- Iced Lavender Cream Oatmilk Matcha
- 120B x1
- Mon 1 apr 2025, table 1, 14:00

Buttons:

- Order Now (Modal)
- Order Now (Product List)

When clicking the Order Now button, the website displays Payment method, Payment information, and Order. The user must input their information and coupon, then click the verify button to use the coupon. Then click the “Order Now” button.

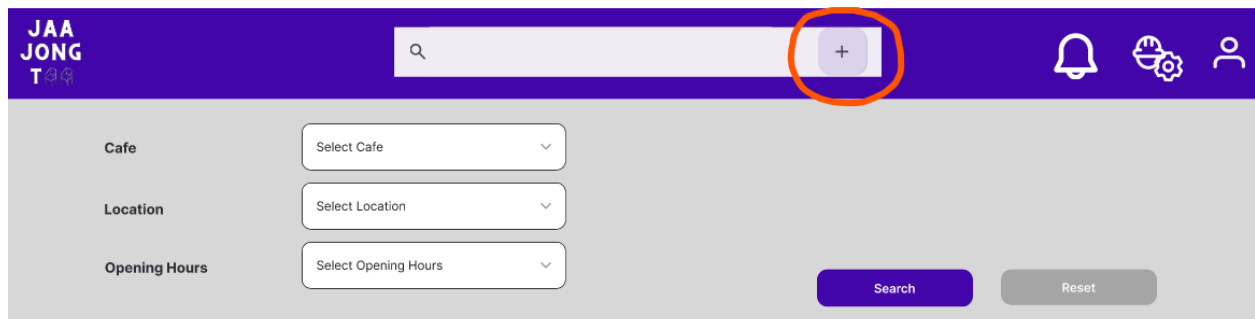


After clicking the Order Now button, the website displays the order's receipt consists of the order number, bank transfer, payment date and time, Payment method, customer name, and the user's order.

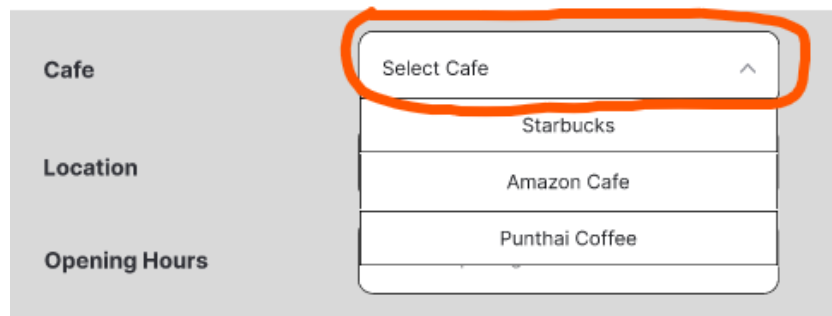
Test Case

Field	Description
Test_ID	TC_P3_001
Test_Name	Search the shop with a filter
Test_Procedure	1. Click the + sign on the search bar. 2. Choose the shop branch. 3. Choose the shop location. 4. Choose the opening hour. 5. Click the “search” button.
Expected_Result	- The search filter should apply immediately after the “search” button is clicked. - The search result should show under the filter bar.

STEP 1: Click the + sign on the search bar.



STEP 2: Choose the shop branch.



STEP 3: Choose the shop location.

Cafe	Select Cafe ▼
Location	Select Location ^
Opening Hours	Near me
	Bangkok
	Nakhon Pathom

STEP 4: Choose the opening hour.

Cafe	Starbucks ▼
Location	Near me ▼
Opening Hours	Select Opening Hours ^
	Standard Hours
	Open 24 Hours

STEP 5: Click the “search” button.

JAA JONG
T 000

🔍 +

🔔 ⚙️ 👤

Cafe	Starbucks ▼
Location	Near me ▼
Opening Hours	Standard Hours ▼

Search Reset

CHECK - By looking at the screen after the searching process

JAA JONG
TAP



Cafe

Starbucks

▼

Location

Near me

▼

Opening Hours

Standard Hours

▼

Search

Reset

- 

Mahidol University - Salaya Campus

 999 Phutthamonthon 4 Road , Salaya , Nakhon
Patthom 73170

- Monday 07:00 to 19:00
 - Tuesday 07:00 to 19:00
 - Wednesday 07:00 to 19:00
 - Thursday 07:00 to 19:00
 - Friday 07:00 to 19:00
 - Saturday 08:00 to 17:00
 - Sunday 08:00 to 17:00
- 

The Fourth Putthamonthon Sai 4

 Phutthamonthon Sai 4 Road ,Kratumlom,Sam
Phran,Nakornpathom

- Monday 07:00 to 21:00
 - Tuesday 07:00 to 21:00
 - Wednesday 07:00 to 21:00
 - Thursday 07:00 to 21:00
 - Friday 07:00 to 21:00
 - Saturday Closed
 - Sunday Closed
- 

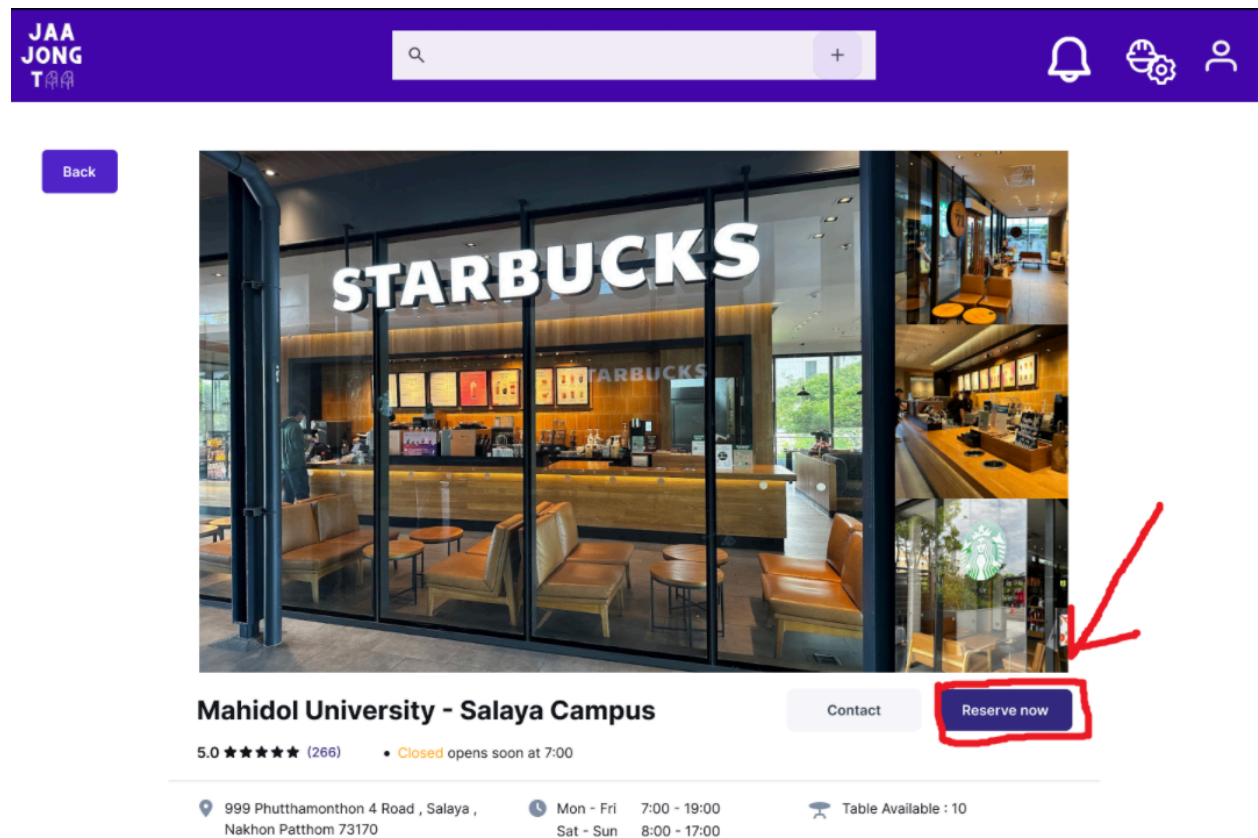
Central Plaza Salaya



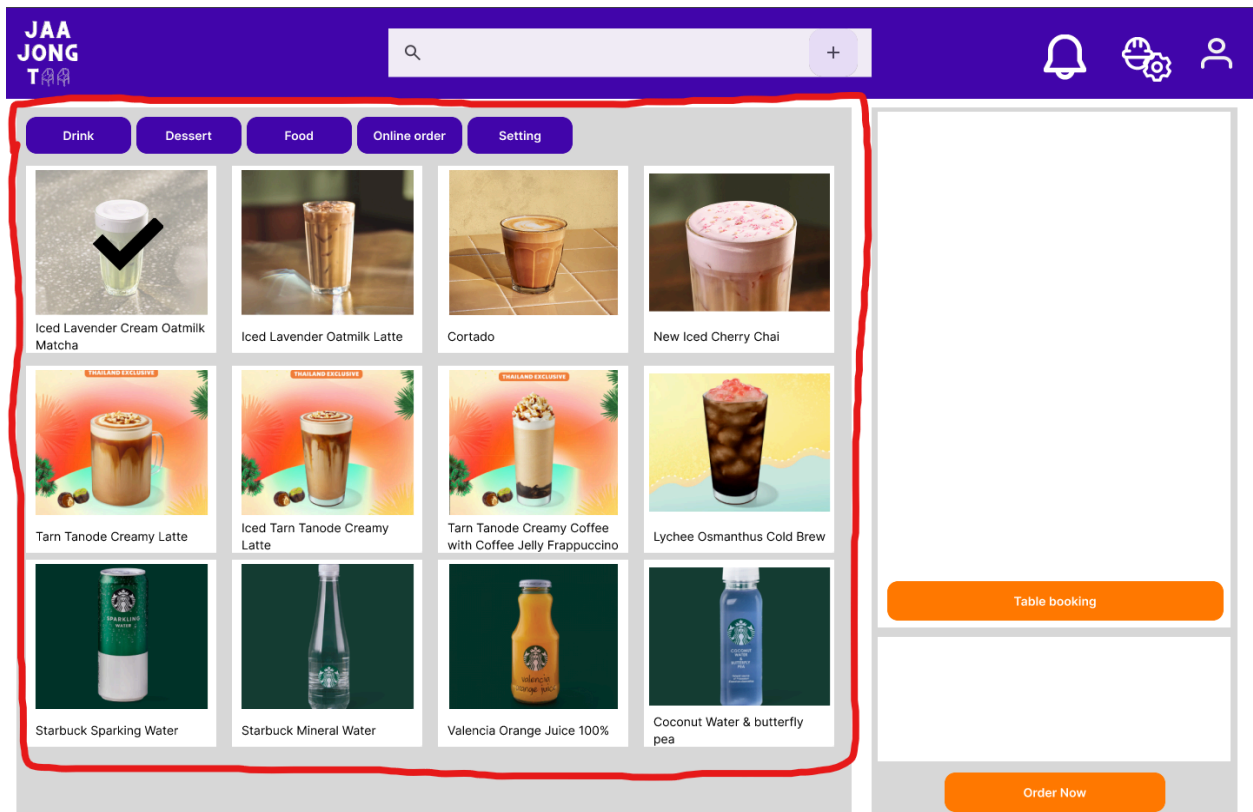
- Monday 07:00 to 21:00
 - Tuesday 07:00 to 21:00

Field	Description
Test_ID	TC_P4_001
Test_Name	Successful Ordering and Seat Reservation
Test_Procedure	1. Click the “Reserve now” button. 2. Select food and drink. 3. Click the “Table booking” button. 4. Select date & time. 5. Click continue 6. Click the “Order Now” button.
Expected_Result	- The order and seat reservation should be sent to the shop staff(Should be able to check via the staff page).

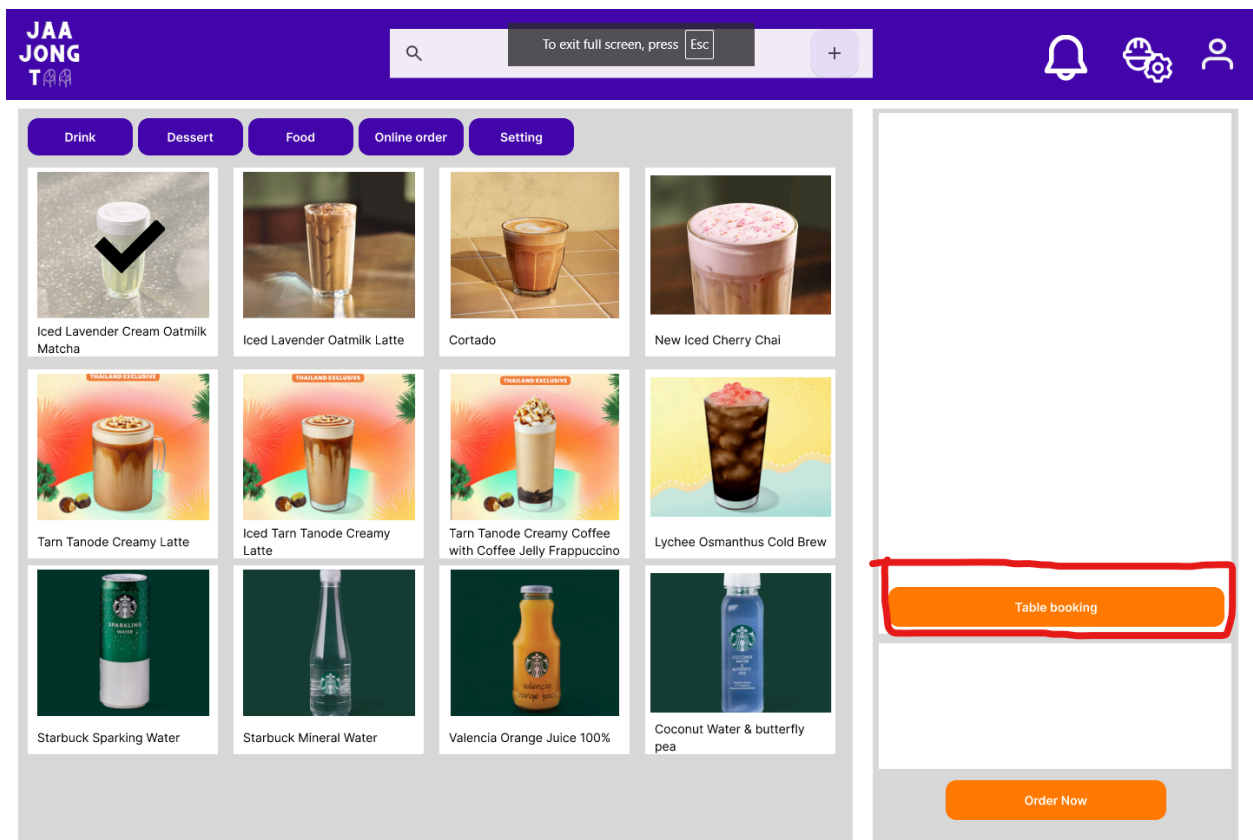
STEP 1: Click the “Reserve now” button.



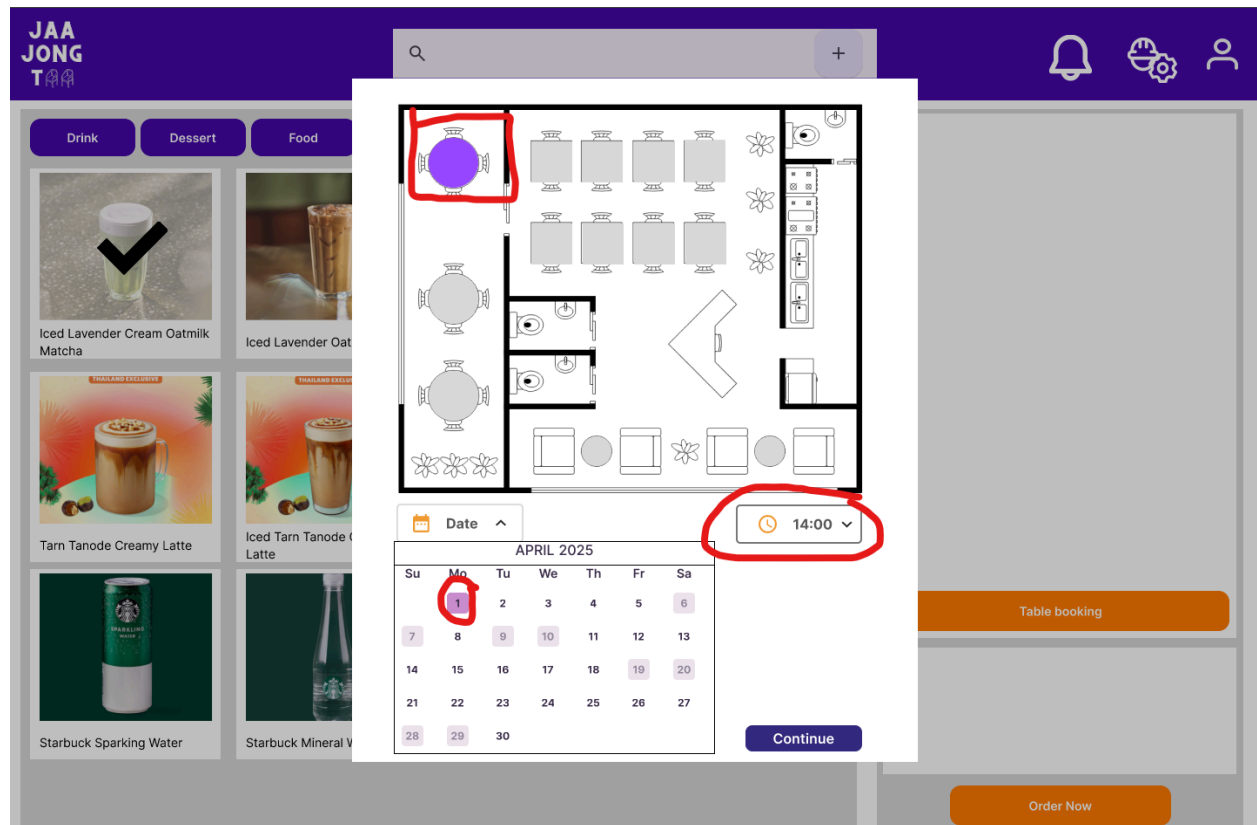
STEP 2: Select food and drink.



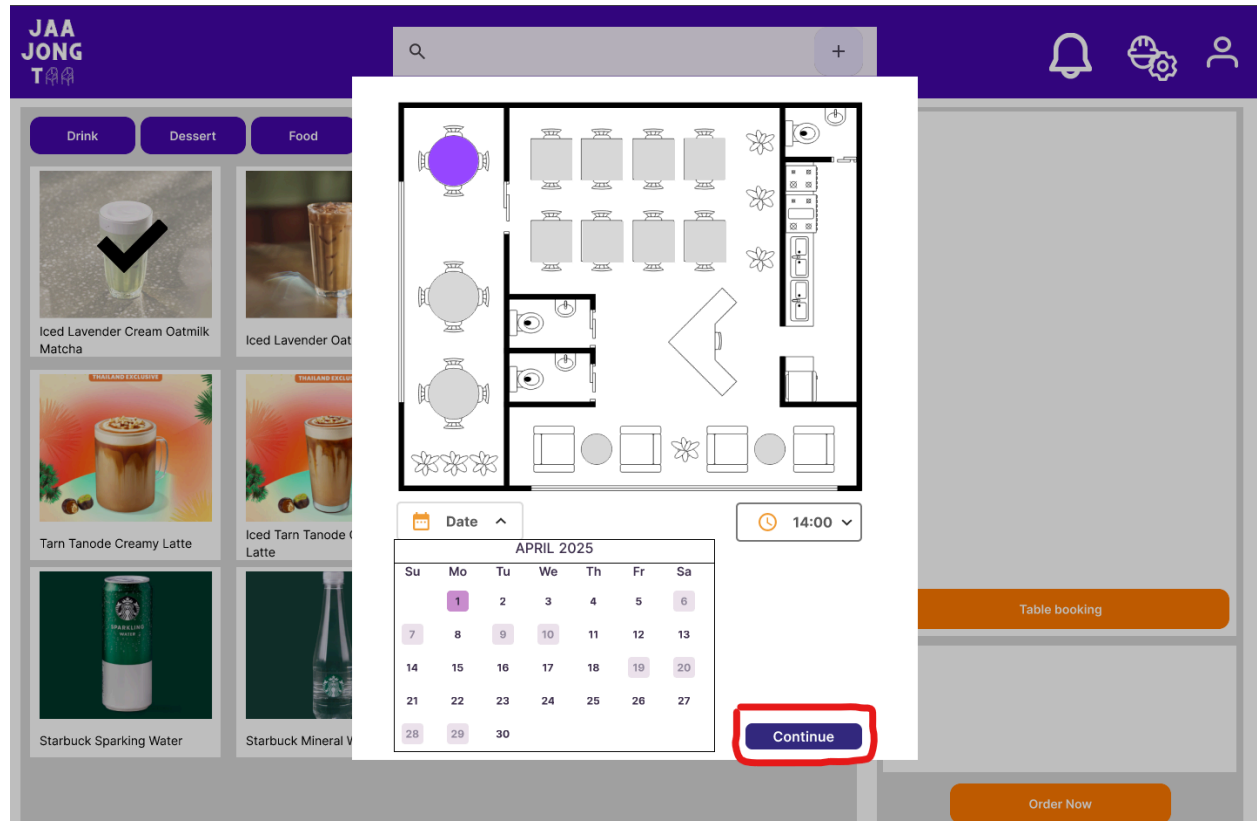
STEP 3: Click the “Table booking” button.



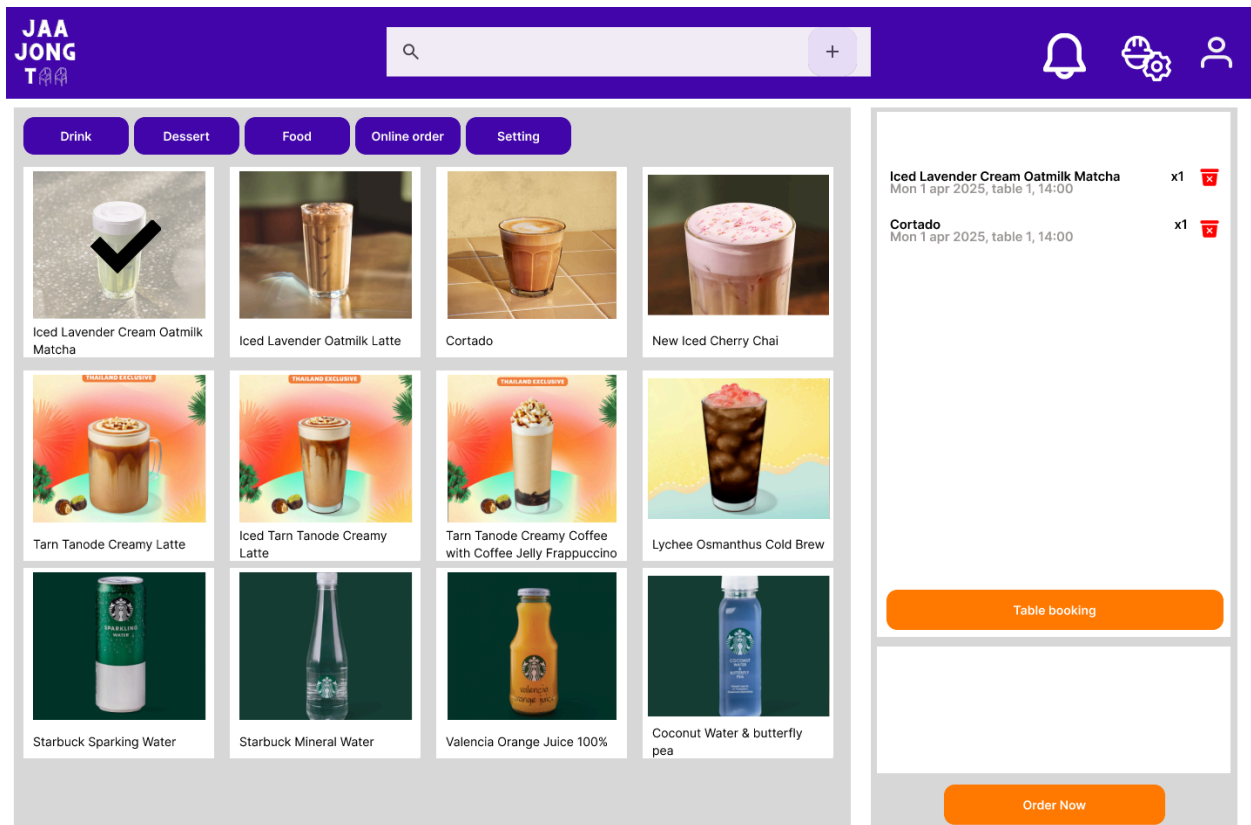
STEP 4: Select date & time.



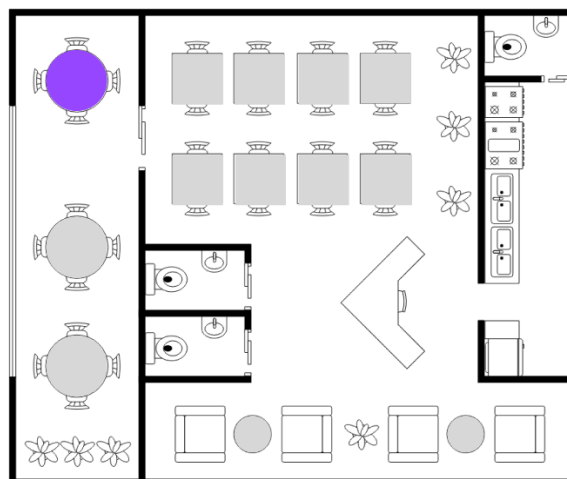
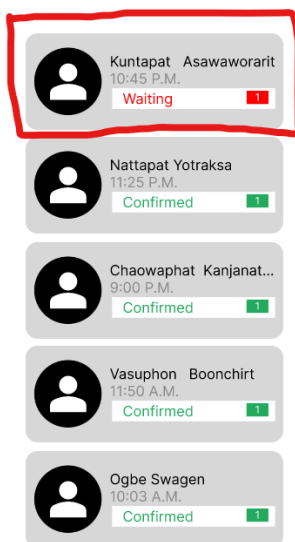
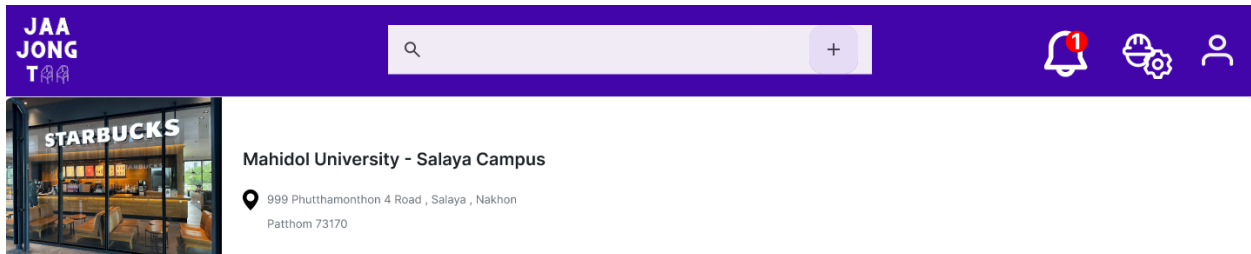
STEP 5: Click continue



STEP 6: Click the “Order Now” button.

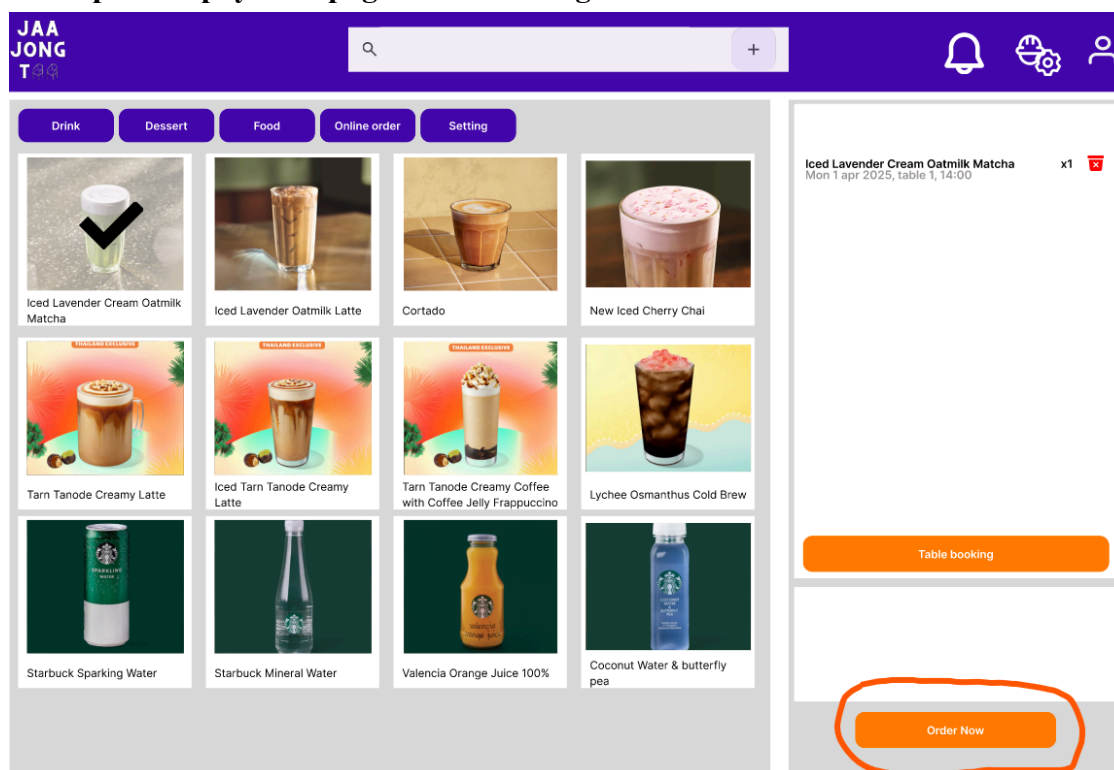


CHECK - By logging in to the staff page and checking the order status

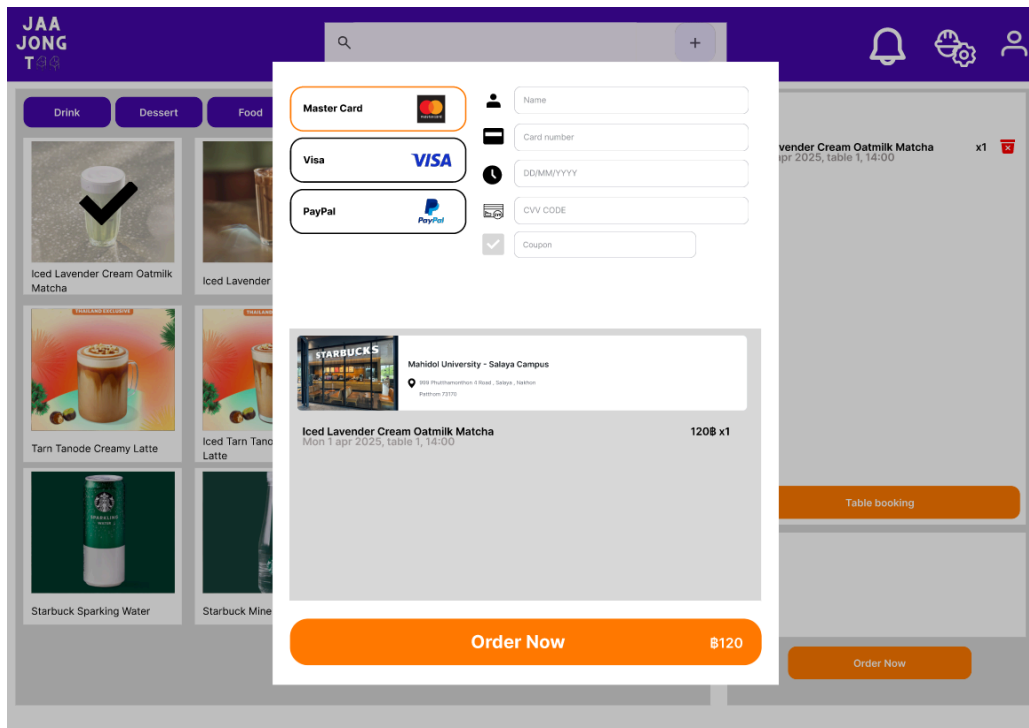


Field	Description
Test_ID	TC_P6_001
Test_Name	Successful Payment with Valid Payment Info and Coupon - Expecting receipt
Test_Procedure	1. Open the payment page after selecting an order. 2. Choose payment method (in this case, credit card). 3. Enter valid payment details (e.g., valid card number, expiration, CVV). 4. Enter a valid coupon code ("NEWUSER30"). 5. Click the "Order Now" button. 6. Get the receipt.
Expected_Result	- Payment is processed successfully. - Coupon is applied, and the final price is discounted. - The receipt is shown to the user.

STEP 1: Open the payment page after selecting an order.



STEP 2: Choose payment method (in this case, credit card).



STEP 3: Enter valid payment details (e.g., valid card number, expiration, CVV).

	Ogbe Swagen
	5333 6195 0371 5240
	12/07/2029
	668
	Coupon

STEP 4: Enter a valid coupon code ("NEWUSER30").

The screenshot shows the JONG TAPP app interface. On the left, there's a menu with categories: Drink, Dessert, and Food. The 'Drink' category is selected, showing items like 'Iced Lavender Cream Oatmilk Matcha', 'Tarn Tanode Creamy Latte', and 'Starbuck Sparkling Water'. The 'Iced Lavender Cream Oatmilk Matcha' item is highlighted with a black checkmark. In the center, there's a checkout form with fields for payment method (Master Card, Visa, PayPal), name (Ogbe Swagen), card number (5333 6195 0371 5240), expiration date (12/07/2029), and a coupon code field. The coupon code 'NEWUSER30' is entered and highlighted with a red circle. Below the coupon code field, there's a green checkmark icon. On the right, there's a summary of the order: 'Iced Lavender Cream Oatmilk Matcha' for 120฿ x1, with a discount of 30฿ x1. The total amount is 90฿. At the bottom, there's an orange 'Order Now' button.

Master Card

Visa

PayPal

Ogbe Swagen

5333 6195 0371 5240

12/07/2029

668

NEWUSER30

STARBUCKS Mahidol University - Salaya Campus
999 Phutthamonthon 4 Road, Salaya, Nakhon Pathom 73170

Iced Lavender Cream Oatmilk Matcha 120฿ x1
Mon 1 apr 2025, table 1, 14:00

Discount 30฿ x1
New user coupon

Order Now 90฿

STEP 5

This screenshot is identical to the one in Step 4, showing the same checkout process. The 'Order Now' button at the bottom is highlighted with a red circle.

Master Card

Visa

PayPal

Ogbe Swagen

5333 6195 0371 5240

12/07/2029

668

NEWUSER30

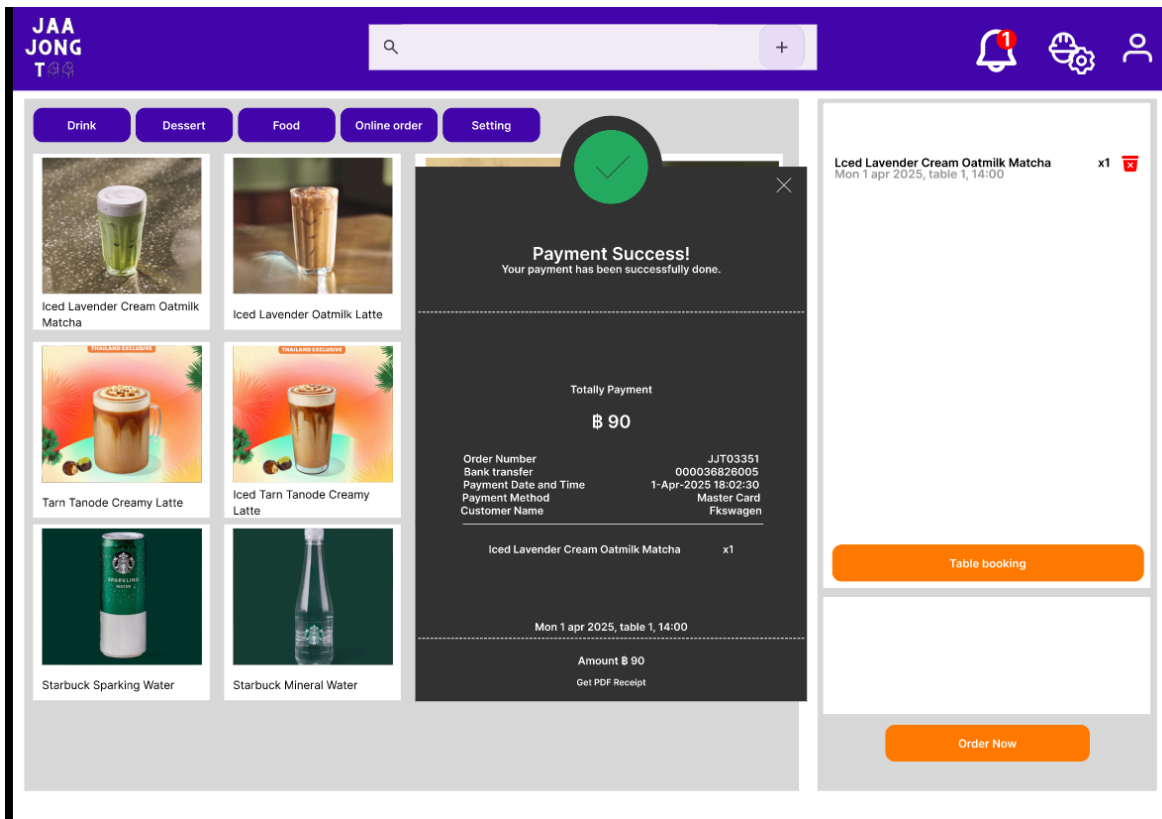
STARBUCKS Mahidol University - Salaya Campus
999 Phutthamonthon 4 Road, Salaya, Nakhon Pathom 73170

Iced Lavender Cream Oatmilk Matcha 120฿ x1
Mon 1 apr 2025, table 1, 14:00

Discount 30฿ x1
New user coupon

Order Now 90฿

STEP 6



CHECK - Look at the receipt after the payment has been made. And the order status has been sent to the customer.

